# 1 Java Language Changes Summary

The following tables summarize new Java language features since Java SE 9.

## New Language Features

The following table lists the new features in the Java language since Java SE 9.

***Table 1-1 Java Language Changes Summary***

| Release | New Language Features |
|---|---|
| 9 | Java Platform Module System (JSR 376): see Project Jigsaw on OpenJDK<br>Milling Project Coin (JEP 213): see Java Language Changes for Java SE 9<br>Small Enhancements to the Java Programming Language (JSR 334): see Java Language Changes for Java SE 9 |
| 10 | Local Variable Type Inference (JEP 286) |
| 11 | Local-Variable Syntax for Lambda Parameters (JEP 323): see Local Variable Type Inference |
| 14 | Switch Expressions and Statements (JEP 361) |
| 15 | Text Blocks (JEP 378) |
| 16 | Record Classes (JEP 395)<br>Pattern Matching with instanceof (JEP 394) |
| 17 | Sealed Classes (JEP 409) |
| 21 | Record Patterns (JEP 440)<br>Pattern Matching with switch (JEP 441) |
| 22 | Unnamed Variables and Patterns (JEP 456) |
| 25 | Module Import Declarations (JEP 511)<br>Compact Source Files and Instance main Methods (JEP 512)<br>Flexible Constructor Bodies (JEP 513) |

## Permanent and Preview Language Features

The following table lists, for each release since Java SE 9, which language features became permanent and which language features were in preview. See Preview Features for information about preview features and how they can become permanent.

***Table 1-2 Java Language Changes Summary***

| Release | Permanent Features | Preview Features |
|---|---|---|
| 25 | Module Import Declarations (JEP 511)<br><br>Compact Source Files and Instance main Methods (JEP 512)<br><br>Flexible Constructor Bodies (JEP 513) | Primitive Types in Patterns, instanceof, and switch (JEP 507): see:<br><br>• When Clauses<br>• Type Patterns with Primitive Types<br>• Primitive Types in Record Patterns<br>• Primitive Values in switch Expressions and Statements<br>• Safe Casting with instanceof and switch |
| 24 | | Module Import Statements (JEP 494)<br><br>Primitive Types in Patterns, instanceof, and switch (JEP 488): see:<br><br>• When Clauses<br>• Type Patterns with Primitive Types<br>• Primitive Types in Record Patterns<br>• Primitive Values in switch Expressions and Statements<br>• Safe Casting with instanceof and switch<br><br>Flexible Constructor Bodies (JEP 482)<br><br>Simple Source Files and Instance Main Methods (JEP 495) |
| 23 | | Module Import Declarations (JEP 476)<br><br>Primitive Types in Patterns, instanceof, and switch (JEP 455): see:<br><br>Implicitly Declared Classes and Instance Main Methods (JEP 477)<br><br>Flexible Constructor Bodies (JEP 482) |

| Release | Permanent Features | Preview Features |
| --- | --- | --- |
| 22 | Unnamed Variables and Patterns (JEP 456) | Statements Before super(...) (JEP 447)<br><br>String Templates (JEP 459)<br><br>Implicitly Declared Classes and Instance Main Methods (JEP 463) |
| 21 | Record Patterns (JEP 440)<br><br>Pattern Matching with switch (JEP 441) | String Templates (JEP 430)<br><br>Unnamed Patterns and Variables (JEP 443)<br><br>Unnamed Classes and Instance Main Methods (JEP 445) |
| 20 | | Pattern Matching for switch Expressions and Statements (JEP 433)<br><br>Record Patterns (JEP 432) |
| 19 | | Pattern Matching for switch Expressions and Statements (JEP 427)<br><br>Record Patterns (JEP 405) |
| 18 | | Pattern Matching for switch Expressions and Statements (JEP 420) |
| 17 | Sealed Classes (JEP 409) | Pattern Matching for switch Expressions and Statements (JEP 406) |
| 16 | Record Classes (JEP 395)<br><br>Pattern Matching with instanceof (JEP 394) | Sealed Classes (JEP 397) |
| 15 | Text Blocks (JEP 378) | Sealed Classes (JEP 36)<br><br>Record Classes (JEP 384)<br><br>Pattern Matching for instanceof (JEP 375) |
| 14 | Switch Expressions and Statements (JEP 361) | Pattern Matching for the instanceof Operator (JEP 305)<br><br>Records (JEP 359)<br><br>Text Blocks (JEP 368) |
| 13 | | Text Blocks (JEP 355), see Programmer's Guide to Text Blocks<br><br>Switch Expressions (JEP 354) |

| Release | Permanent Features | Preview Features |
|---|---|---|
| 12 | | Switch Expressions (JEP 325) |
| 11 | Local-Variable Syntax for Lambda Parameters (JEP 323): see Local Variable Type Inference | |
| 10 | Local Variable Type Inference (JEP 286) | |
| 9 | Java Platform Module System (JSR 376): see Project Jigsaw on OpenJDK<br><br>Milling Project Coin (JEP 213): see Java Language Changes for Java SE 9<br><br>Small Enhancements to the Java Programming Language (JSR 334): see Java Language Changes for Java SE 9 | |

## Evolution of Language Features

The following tables list the changes made to the Java language since Java SE 9. The first column specifies the feature while the subsequent columns specify a Java SE release. An icon specifies whether the feature was made permanent, withdrawn for the specified release, or available as a preview feature:

- P: Permanent feature
- W: The feature was withdrawn
- 1, 2, 3, 4: First, second, third, or fourth preview, respectively

See Preview Features for information about preview features and how they can become permanent or be withdrawn.

***Table 1-3 Java Language Changes in Java SE 25***

| Feature | 25 |
|---|---|
| Module Import Declarations | P JEP 511 |
| Primitive Types in Patterns, instanceof, and switch | 3 JEP 507 |
| Flexible Constructor Bodies | P JEP 513 |
| Compact Source Files and Instance main Methods | P JEP 512 |

***Table 1-4 Java Language Changes from Java SE 24 to Java SE 17***

| Feature | 24 | 23 | 22 | 21 | 20 |
|---|---|---|---|---|---|
| Module Import Declarations | 2 JEP 494 | 1 JEP 476 | | | |
| Primitive Types in Patterns, instanceof, and switch | 2 JEP 488 | 1 JEP 455 | | | |
| Flexible Constructor Bodies | 3 JEP 492 | 2 JEP 482 | 1 JEP 447 | | |
| Simple Source Files and Instance Main Methods | 4 JEP 495 | 3 JEP 477 | 2 JEP 463 | 1 JEP 445 | |
| Unnamed Variables and Patterns | | | P JEP 456 | 1 JEP 443 | |
| String Templates | | W | 2 JEP 459 | 1 JEP 430 | |
| Record Patterns | | | | P JEP 440 | 2 JEP 432 |
| Pattern Matching with switch | | | | P JEP 441 | 4 JEP 433 |
| Sealed Classes | | | | | |

*Table 1-5 Java Language Changes from Java SE 16 to Java SE 9*

| Feature | 16 | 15 | 14 | 13 | 12 |
|---|---|---|---|---|---|
| Sealed Classes | 2 JEP 397 | 1 JEP 360 | | | |
| Record Classes | P JEP 395 | 2 JEP 384 | 1 JEP 359 | | |
| Pattern Matching with instanceof | P JEP 394 | 2 JEP 375 | 1 JEP 305 | | |
| Text Blocks | | P JEP 378 | 2 JEP 368 | 1 JEP 355 | |
| Switch Expressions | | | P JEP 361 | 2 JEP 354 | 1 JEP 325 |

| Feature | 16 | 15 | 14 | 13 | 12 |
| --- | --- | --- | --- | --- | --- |
| and Statements | | | | | |
| Local-Variable Syntax for Lambda Parameters: see Local Variable Type Inference | | | | | |
| Local Variable Type Inference | | | | | |
| Java Platform Module System: see Project Jigsaw on OpenJDK | | | | | |
| Milling Project Coin: see Java Language Changes for Java SE 9 | | | | | |
| Small Enhancements to the Java Programming Language: see Java Language Changes for Java SE 9 | | | | | |